Lab 01 — Requirements Engineering & UML Use-Case Modelling

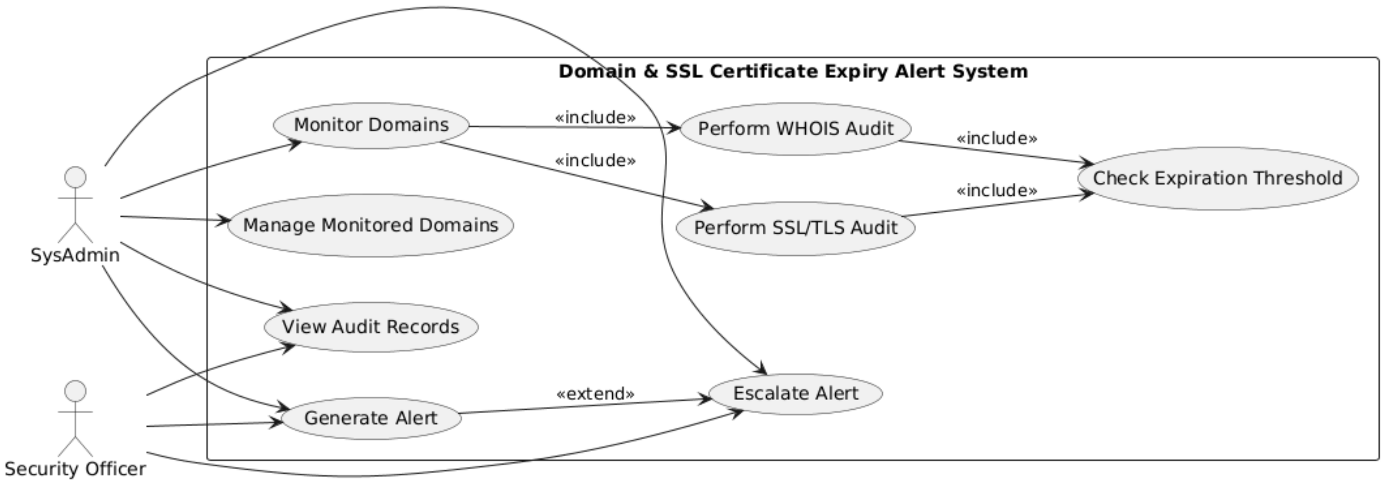
Problem Statement #47: Domain & SSL Certificate Expiry Alert System

1. Requirements Specification

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates, and trigger alerts at 30, 15, and 3 days before expiry.	High	Pass: An alert is queued when a certificate is expiring in 30, 15, or 3 days. Fail: An expiring certificate reaches a configured threshold without an alert.	Prevents unexpected SSL/TLS certificate expiration and service disruption.
FR-002	Functional	The system shall perform daily WHOIS registration audits for all monitored domains and retrieve their domain registration expiration dates.	High	Pass: WHOIS registration expiry information is retrieved and recorded for every monitored domain. Fail: A monitored domain with available WHOIS information has no recorded expiry information.	Prevents domain registration expiry from causing loss of domain availability.
FR-003	Functional	The system shall compare domain and SSL/TLS certificate expiration dates against the configured expiration thresholds.	High	Pass: The system correctly identifies whether an expiry date falls within a configured alert threshold. Fail: An expiry within a configured threshold is incorrectly classified as safe.	Ensures that alerts are generated at the appropriate time before expiration.
FR-004	Functional	The system shall send alerts to the appropriate SysAdmin or Security Officer according to the configured escalation ladder.	High	Pass: Alerts are delivered to the appropriate stakeholder according to the escalation level. Fail: A required alert is generated but is not delivered or escalated.	Ensures that responsible personnel are notified and can take corrective action.
FR-005	Functional	The system shall maintain an audit record containing the monitored domain, audit date, expiration information, alert status, and escalation status.	Medium	Pass: Each completed audit has a retrievable record containing all required information. Fail: An audit record is missing or does not contain the required information.	Provides traceability of monitoring activities, alerts, and escalations.
NFR-001	Performance	The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.	High	Pass: Benchmarking confirms that 1,000 domains and SSL endpoints are scanned in under 3 minutes. Fail: The scan takes 3 minutes or longer.	Ensures that daily monitoring completes within an acceptable operational time.
NFR-002	Security	The system shall protect monitoring and audit information from unauthorized access and restrict alert-management operations to authorized users.	High	Pass: Unauthorized users cannot access audit information or modify alert and escalation settings. Fail: An unauthorized user can access or modify protected information.	Protects monitoring data and prevents unauthorized changes to alert configurations.

2. UML Use-Case Diagram

Domain & SSL Certificate Expiry Alert System



Actors: SysAdmin and Security Officer. The model includes domain monitoring, WHOIS and SSL/TLS audits, expiration-threshold checking, alert generation, alert escalation, domain management, and audit-record viewing. The diagram uses «include» for mandatory audit/check behavior and «extend» for escalation behavior.

3. Use-Case Flow Specification

Use Case: Monitor Domain and SSL/TLS Certificate Expiration

Use Case ID: UC-001

Primary Actors: SysAdmin; Security Officer

Goal: To automatically monitor domain registration and SSL/TLS certificate expiration dates and alert responsible personnel before expiration.

Preconditions

1. The monitoring system is operational.
2. The domains to be monitored are registered in the system.
3. The system has network access required to perform WHOIS audits and SSL/TLS handshakes.
4. Alert recipients and the escalation ladder are configured.

Postconditions

1. WHOIS and SSL/TLS audit results are recorded.
2. Domain registration and SSL/TLS certificate expiration dates are stored.
3. Required alerts are generated when an expiration threshold is reached.
4. Alert and escalation activities are recorded in the audit records.

Main Success Scenario

1. The monitoring system initiates the scheduled daily monitoring process.
2. The system retrieves the list of monitored domains.
3. The system performs a WHOIS registration audit for each monitored domain.
4. The system performs an SSL/TLS handshake against each monitored domain.
5. The system extracts the domain registration and SSL/TLS certificate expiration dates.
6. The system compares the expiration dates against the configured thresholds.
7. The system identifies domains or certificates approaching expiration.
8. If an expiration date is 30, 15, or 3 days away, the system generates an alert.
9. The system sends the alert to the appropriate SysAdmin or Security Officer.
10. If escalation is required, the system escalates the alert according to the configured escalation ladder.
11. The system records the audit results, alert status, and escalation status.
12. The monitoring process completes successfully.

Alternate Flow: Audit Failure

1. During the WHOIS or SSL/TLS audit, the system cannot reach a domain or retrieve the required information.
2. The system records the audit as unsuccessful.
3. The system records the reason for the audit failure.
4. The system continues the monitoring process for the remaining domains.
5. The failed audit is recorded for review by the SysAdmin or Security Officer.